MTH632 Quiz=1

Topic (1-12)

Due Date (19-24) Dec2021

Which one cancel because out of syllabus

Again dates (1-3) Feb2022 but quiz out of syllabus

- MTH632 MID TERM SOLVED 13june2019 pg12
- Quiz MTH632(ZE+BA File)(2) 13june2019 pg10
- Mth632 quiz 1 19june2020 pg10
- Mth632 quiz1 solved (Module 29 To 37) 25july2021 pg33
- Mth632 quiz 1 solved 25may2021 pg40
- Mth632 quiz1 repeted mcqs 25may2021 pg40 unsolve
- Mth632 Quiz no.1 by s khan academy 25may2021 pg12
- Quiz 1 mth632 19dec2021 pg18
- MTH 632 QUIZ 1 (BY KING) pg10
- MTH632 Quiz1.by(KING) pg18
- Mth632 current Quiz 1Dec 2k21 22dec2021 pg68
- Mth632 current Quiz 1Dec 2k21-1 22dec2021 pg68 unsolve
- Mth632 Quiz#1 1_Feb_2022 pg60
- MTH632 BY S KHAN ACADEMY pg13 in this file only quiz1
- Mth632 quiz in 4 groups no limit pg 1000+ quiz

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 1 of 10 (Start time: 05:56:24 PM, 13 June 2019)

Total Marks: 1

The derivative of f at z_0 i.e. $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ *provided the limit exists. The function f must be defined is a neighbourhood of* _____.

Select the correct option

Reload Math Equations

○	z
●	z_0

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Question # 2 of 10 (Start time: 05:57:15 PM, 13 June 2019)

Total Marks: 1

If a function f(z) = u(x,y) + iv(x,y) is analytic in a domain D, then its component functions u and v are _____ in D.

Select the correct option

- ○ equal
- ○ steady
- ● harmonic ✓
- ○ homogeneous

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 3 of 10 (Start time: 05:57:27 PM, 13 June 2019)

Total Marks: 1

If Cauchy – Riemann equations are not satisfied at a point z_0 then function is not _____ at z_0.

Select the correct option

Reload Math Equations

○	integrable
●	differentiable ✓

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Question # 9 of 10 (Start time: 10:13:26 PM, 13 June 2019) Total Marks

If f(z) = x + ay + i(bx + cy) is analytic, then a, b, c equals to _____.

Select the correct option

- a= 1 and c= -b
- a = b = c = 1
- b = 1 and a = -c
- ● c=1 and a= -b ✓

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MTH632:Quiz#1

Question # 5 of 10 (Start time: 05:58:16 PM, 13 June 2019)

In the complex valued function z = x - iy, the value of $U_z =$ _____.

Select the correct option

○	-1
○	0
●	1
○	2

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MTH632:Quiz#1

Question # 7 of 10 (Start time: 05:58:58 PM, 13 June 2019)

In the complex valued function z = x - iy, the value of $V_z =$ _____.

Select the correct option

○	-1
●	0
○	1
○	2

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 9 of 10 (Start time: 05:59:28 PM, 13 June 2019)

Total Marks: 1

The Cauchy - Riemann equations on a pair of real - valued functions of two real variables $u(x, y)$ and $v(x, y)$ are $U_x = V_y$ and _____.

Select the correct option

Reload Math Equations

- $U_y = -V_y$
- $U_x = -V_x$
- $U_y = -V_x$ ✓
- $U_x = -V_y$

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 10 of 10 (Start time: 05:59:52 PM, 13 June 2019)

Total Marks: 1

A function f(z) is analytic function if _____ part f(z) is analytic.

Select the correct option

- (a) real
- (b) imaginary
- (c) both (a) and (b).
- (d) none of these.

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M11632 Quiz # 1

Question # 9 of 10 (Start time: 07:48:47 PM, 25 May 2021)

The fifth roots of -1 are exactly

Select the correct option

- cos(± π/10), cos(±3π/10), and -1
- cos(± π/5), cos(±3π/5), and -1 ✓
- cos(π/5), cos(3π/5), cos((-3π)/5) and cos((-π)/5)
- none

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MTH632 Quiz # 1

Question # 8 of 10 (Start time: 08:40:08 PM, 25 May 2021)

If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(i)) =$

_____

Select the correct option

- i^2
- i^3
- -1
- i ✓

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MTH632 Quiz # 1

Question # 3 of 10 (Start time: 07:46:27 PM, 25 May 2021)

If $f(x) = x^3$, $g(x) = x + 2$, then $g(f(5)) =$

_____

Select the correct option

○	130
○	7
○	125
●	127

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Question # 4 of 10 (Start time: 08:19:52 PM, 25 May 2021)

If we write $\Delta z = z - z_0$, then the equation $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ can be expressed as $f'(z_0) =$ _____.

Select the correct option

- $\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z)}{\Delta z}$
- $\lim_{\Delta z \to 0} \frac{f(\Delta z) - f(z_0)}{\Delta z}$
- $\lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$
- ✓ $\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$

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MTH632 Quiz # 1

Question # 1 of 10 (Start time: 07:44:14 PM, 25 May 2021)

If $f(z) = z^3$, $g(z) = z + 2$, then $g(f(4)) =$

_____

Select the correct option

- 4
- 66
- 64
- 16

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MTH632:Quiz # 1

Question # 10 of 10 (Start time: 08:27:07 AM, 25 May 2021)

In general, $1^{a}(1/n)$ is

Select the correct option

O	$\exp\left(i\frac{2k\pi}{n}\right),\ k=0,1,\ldots,n-1$
O	$\exp\left(i\frac{2\pi}{n}\right)$

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Question # 6 of 10 (Start time: 08:23:17 AM, 25 May 2021)

Total Marks: 1

The distance between two complex numbers z_1 and z_2 is

Select the correct option

Reload Math Equations

O	$\lvert z_1 - z_2 \rvert$
O	$\lvert z_1 + z_2 \rvert$

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Question # 8 of 10 (Start time: 08:25:18 AM, 25 May 2021)

Total Marks

A closed disc with center z_0 and radius ε is defined by

Select the correct option

Reload Math Equations

O	$\lvert z_1 - z_0 \rvert \geqslant \varepsilon$
O	$\lvert z_1 - z_0 \rvert \leqslant \varepsilon$

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Question # 2 of 10 (Start time: 08:19:04 AM, 25 May 2021)

If $f(x) = x^2$, $g(x) = \frac{1}{x}$, then $g(f(3)) =$ _____.

Select the correct option

O	1/3
O	3
O	1/9
O	9

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Question # 4 of 10 (Start time: 08:21:20 AM, 25 May 2021)

The three cube roots of the number -8i are

Select the correct option

○ $2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1,2$

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○ $2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1$

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Question # 2 of 5 (Start time: 03:38:19 PM, 21 December 2021)

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z)g(z)) =$ _____.

Select the correct option

- ○ $A + B$
- ○ $A.B$
- ○ $A - B$
- ○ $A \pm B$

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Question # 4 of 5 (Start time: 03:55:12 PM, 22 December 2021)

What is the modulus of the complex number 6+0i?

Select the correct option

- -6
- 6
- 0
- None of these

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Quiz Start Time: 08:24 PM, 21 December 2021

Question # 3 of 5 (Start time: 08:25:55 PM, 21 December 2021)

Total Marks: 1

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z) \pm g(z)) =$ _____.

Select the correct option

Reload Math Equations

- $A \pm B$
- $A - B$
- A
- $A + B$

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Question # 3 of 5 (Start time: 11:37:10 AM, 19 December 2021) **Total Marks: 1**

$$|z_1 - z_2| =$$

Select the correct option

Reload Math Equations

○	$> |z_1| + |z_2|$
○	$> z_1 + z_2$
○	$\leqslant z_1 + z_2$
◉	$\leqslant |z_1| + |z_2|$

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sec(s)

MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) Quiz Start Time: 02:06 AM, 21 December 2021

Question # 4 of 5 (Start time: 02:09:25 AM, 21 December 2021) **Total Marks:** 1

The complex number -1-i has exponential form

Select the correct option

Reload Math Equations

- (●) $\sqrt{2}\left[i\left(-\frac{3\pi}{4}\right)\right]$
- ( ) $\sqrt{2}\left[i\left(-\frac{3\pi}{2}\right)\right]$

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Question # 5 of 5 (Start time: 11:38:51 AM, 19 December 2021)

$$Consider \lim_{z \to z_0} f(z) = A \text{ and } \lim_{z \to z_0} g(z) = B. \text{ Then } \lim_{z \to z_0} \frac{f(z)}{g(z)} = _____.$$

Select the correct option

○	$A - B$
◉	$\frac{A}{B},\ B \neq 0$ ✓
○	$A + B$
○	$A \,.\, B$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 07:47:31 PM, 21 December 2021)

If $-8i = 8\exp[i(-\pi/2+2k\pi)]$, then the root of

$$(-8i)^{1/3}$$

for k=0 is _____

Select the correct option

○	$2(\sqrt{3}+i)$
○	$\sqrt{3}-i$
○	$\sqrt{3}+i$
○	$2(\sqrt{3}-i)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) Quiz Start Time: 02:06 AM, 21 December 2021

Question # 3 of 5 (Start time: 02:08:35 AM, 21 December 2021) **Total Marks: 1**

Let z = 3, then arg(z) = _____.

Select the correct option

Reload Math Equations

◉	270 degree
○	60 degree
○	90 degree
○	0 degree

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 07:46:35 PM, 21 December 2021)

Domain of the function

$$f(z) = \frac{1}{z^2 - 1}$$

is _____

Select the correct option

- $z \neq \pm i$
- $z \neq \pm 1$ ✓
- $z \neq 0$
- $z \neq \pm 2$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:56:28 PM, 21 December 2021)

If $-8i = 8\exp[\,i(-\pi/2+2k\pi)\,]$, then the root of

$$(-8i)^{1/3}$$

for k=2 is _____

Select the correct option

○	$-\sqrt{3}-i$
○	$-\sqrt{3}+i$
○	$\sqrt{3}-i$
○	$\sqrt{3}+i$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 03:47:57 PM, 21 December 2021)

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y = _____$$

Select the correct option

○	8
○	10
○	9
○	11

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:40:01 PM, 21 December 2021)

Sum of complex numbers $(3 + 5i)$ and $(4 - 3i)$ is _____

Select the correct option

○	$7 + 2i$
○	$1+5i$
○	$7-3i$
○	$7+5i$

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Question # 4 of 5 (Start time: 03:41:02 PM, 21 December 2021) **Total Marks: 1**

The third roots of 2i is

Select the correct option

Reload Math Equations

- $\sqrt{3}\left[i\left(\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k \in \mathbb{Z}$
- $\sqrt[3]{2}\left[i\left(\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k \in \mathbb{Z}$

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Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 03:38:52 PM, 21 December 2021)

A complex-valued linear transformation could be a composition of _____

Select the correct option

- translation
- magnification
- all of the above
- rotation



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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:37:52 PM, 21 December 2021)

The complex number

$$(4i)^4$$

have _____ number of roots

Select the correct option

○	2
○	1
○	4
○	3

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Question # 2 of 5 (Start time: 03:28:09 PM, 21 December 2021)

Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$, then choose the correct option.

Select the correct option

○	$\lim_{z \to z_0} [f(z) + g(z)] = A + B$
○	$\lim_{z \to z_0} [f(z) + g(z)] = \frac{A}{B}$
○	$\lim_{z \to z_0} [f(z) + g(z)] = A \cdot B$
○	$\lim_{z \to z_0} [f(z) + g(z)] = A - B$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:04:20 AM, 01 February 2022)

The x - axis is called _____.

Select the correct option

○	an imaginary axis
○	a real axis ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 12:05:38 AM, 01 February 2022)

Conjugate of complex number 8 + 3 i is _____.

Select the correct option

○	-8 - 3i
○	-8 + 3i
○	8 + 3i
○	8 - 3i ✔

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 12:23:50 AM, 01 February 2022)

Difference of complex numbers (4 + 2i) and (1 + 8i) is _____.

Select the correct option

○	(3 - 10i)
○	(5 - 6i)
○	(5 - 10i)
○	(3 - 6i) ✓

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Question # 4 of 5 (Start time: 12:25:27 AM, 01 February 2022)

If initial point and terminal points are equal then the curve C is said to be_____.

Select the correct option

- ○ closed curve
- ○ parallel curve
- ○ open curve
- ○ simple curve

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 12:27:02 AM, 01 February 2022)

Product of complex numbers (3 + 2i) and (1+7i) is _____.

Select the correct option

○	3+9i
○	4+9i
○	-11+23i ✓
○	3+14i

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Question # 3 of 5 (Start time: 09:32:10 AM, 01 February 2022)

The solution of the equation $x^2 + x + 1 = 0$ lies in the set R of real numbers.

Select the correct option

○	True
◉	False



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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 08:38:48 PM, 01 February 2022)

A real number x can be displayed as a point _____ on real axis.

Select the correct option

○	(1, x)
○	(0, x)
○	(x, 1)
◉	(x, 0) 

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 08:39:09 PM, 01 February 2022)

$z_1 = (1, 3)$ and $z_2 = (3, 1)$ are _____.

Select the correct option

◉	unequal
○	equal

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 08:41:44 PM, 01 February 2022)

The y - axis is called _____.

Select the correct option

○	a real axis
◉	an imaginary axis ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 08:41:56 PM, 01 February 2022)

The commutative law shows that _____.

Select the correct option

○	$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$
○	$z_1 (z_2 + z_3) = (z_1 z_2 + z_1 z_3)$
◉	$z_1 + z_2 = z_2 + z_1$ ✓
○	$z + w = z$

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Question # 5 of 5 (Start time: 08:42:38 PM, 01 February 2022)

Domain of the function

$$f(z) = \frac{1}{z^2}$$

Is _____.

Select the correct option

○	$z \neq \pm 2$
◉	$z \neq \pm i$
○	$z \neq \pm 1$
○	$z \neq 0$ ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 08:51:02 PM, 01 February 2022)

Let z = 7 i, then Arg(z) = _____.

Select the correct option

○	$\frac{\pi}{3}$
○	$\frac{\pi}{4}$
◉	$\frac{\pi}{2}$ ✓
○	$\frac{\pi}{6}$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 08:51:23 PM, 01 February 2022)

Let z = 1 + i, then Arg(z) = _____.

Select the correct option

○	2π/4
◉	π/4 ✓
○	3π/4
○	4π/4

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 08:51:44 PM, 01 February 2022)

The additive identity of a complex number $z = (x, y)$ is _____.

Select the correct option

○	$(1, 0)$
◉	$(0, 0)$ 

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 08:52:30 PM, 01 February 2022)

The conjugate of 6 + 3i is

Select the correct option

○	-6-3i
◉	6-3i 
○	-6+3i
○	6+3i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 09:05:03 PM, 01 February 2022)

Conjugate of complex number $-7 - 4i$ is _____.

Select the correct option

◉	$-7 + 4i$ ✓
○	$7 - 4i$
○	$7 + 4i$
○	$-7 - 4i$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 09:05:20 PM, 01 February 2022)

What is the modulus of the complex number -6 +8i?

Select the correct option

○	-10
◉	10
○	14
○	6

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:05:45 PM, 01 February 2022)

What is the modulus of the complex number 8i?

Select the correct option

○	-8
○	i
◉	8
○	-i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:06:01 PM, 01 February 2022)

The solution of the equation $x^2 + 1 = 0$ lies in the set of complex numbers.

Select the correct option

◉	True
○	False

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Question # 1 of 5 (Start time: 09:21:18 PM, 01 February 2022)

The polar form of a complex number z is _____.

Select the correct option

○	$r(sec\theta + i\cos ec\theta)$
◉ ✓	$r(cos\theta + i\sin\theta)$
○	$r(tan\theta + i\cot\theta)$
○	$r(sin\theta + i\cos\theta)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 09:22:28 PM, 01 February 2022)

The equation $x^2 - 2x + 3 = 0$ does not have the real roots.

Select the correct option

◉	False
○	True ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:23:50 PM, 01 February 2022)

Let z = 1 + i, then r = _____.

Select the correct option

○	2
○	$\sqrt{3}$
◉	$\sqrt{2}$ ✓
○	1

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:24:07 PM, 01 February 2022)

The additive inverse of (5, 7) is _____.

Select the correct option

○	(5, -7)
○	(-5, 7)
◉	(-5, -7) ✓
○	(5, 7)

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 09:49:40 PM, 01 February 2022)

$$|z_1 + z_2| =$$

Select the correct option

○	$> z_1 + z_2$
○	$> \|z_1\| + \|z_2\|$
◉ ✓	$\leqslant \|z_1\| + \|z_2\|$
○	$\leqslant z_1 + z_2$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 09:50:29 PM, 01 February 2022)

The limit $\lim_{z \to 0} \frac{z}{\bar{z}}$ _____.

Select the correct option

○	exists
◉	does not exists



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Question # 3 of 5 (Start time: 09:50:49 PM, 01 February 2022)

In z = a + bi, if i is replaced by -i, then another complex number obtained is said to be

Select the correct option

○	multiplicative inverse of z
○	prime factor of z
◉	Complex conjugate of z
○	additive inverse of z

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:51:05 PM, 01 February 2022)

$If\ z = a + ib\ is\ a\ complex\ number,\ then$

Select the correct option

○	None of these
○	✓ $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$
○	$\mathrm{Re}(z) = \frac{z+\bar{z}}{2i}$
○	$\mathrm{Re}(z) = z + \bar{z}$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:52:14 PM, 01 February 2022)

The Jordan Curve Theorem guarantees that a simple closed curve must enclose a region.

Select the correct option

○	Euler curve
○	Simple curve
○	Jordan curve ✓
○	Rotation curve

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Question # 1 of 5 (Start time: 10:23:01 PM, 01 February 2022)

The function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit when _____.

Select the correct option

○	$(x,y) \to (1,0)$
○	$(x,y) \to (1,1)$
○	$(x,y) \to (0,1)$
◉	$(x,y) \to (0,0)$ ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:23:21 PM, 01 February 2022)

The Cartesian form of the complex number $6\text{cis}(\pi)$ is

Select the correct option

○	-6π
○	6π
○	6
◉	-6

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 10:23:32 PM, 01 February 2022)

A curve is called _____ if it does not cross itself.

Select the correct option

○	open curve
○	parallel curve
○	closed curve
◉	simple curve 

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 10:24:14 PM, 01 February 2022)

A complex power function is a function of the form _____, where

α is a complex constant.

Select the correct option

◉	$f(z) = z^{\alpha}$ ✓
○	$f(\alpha) = z^{\alpha}$
○	$f(\alpha) = \alpha^{z}$
○	$f(z) = \alpha^{z}$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:24:27 PM, 01 February 2022)

$$(1+i)^2 = _____$$

Select the correct option

○	1+i
○	1−2i
○	1+2i
◉	0+2i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:26:06 PM, 01 February 2022)

If (2, 3) + (a, b) = (0, 0), then (a, b) = _____.

Select the correct option

○	(-2, 3)
◉	(-2, -3)
○	(2, -3)
○	(2, 3)

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:26:23 PM, 01 February 2022)

The representation $a + i\,b$ of a complex number z is due to Gauss.

Select the correct option

◉	True
○	False

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:27:04 PM, 01 February 2022)

$Re(z)$ of $z = (5, 7) + (3, -2)$ is _____.

Select the correct option

◉	8
○	3
○	7
○	5

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:34:37 PM, 01 February 2022)

The root of the equation $6x + 1 = 3$ is _____.

Select the correct option

○	an irrational number.
◉	a rational number 

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:34:51 PM, 01 February 2022)

Mapping is _____ way of visualizing a given function.

Select the correct option

- ○ a numerical
- ○ none of the above
- ○ an analytical
- ◉ a geometrical

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 10:35:02 PM, 01 February 2022)

If z = -1+i, then z expressed in polar form is

Select the correct option

◉	$2cis\left(-\frac{\pi}{4}\right)$
○	$cis\left(-\frac{\pi}{4}\right)$
○	$2cis\left(\frac{3\pi}{4}\right)$
○	$cis\left(\frac{3\pi}{4}\right)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:35:38 PM, 01 February 2022)

$$Let\ z\overline{z} = 64, then\ |z| =$$

Select the correct option

○	-8
◉	8
○	64
○	-64

✓ $z\bar{z} = |z|^2$

$|z| = 8$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:42:18 PM, 01 February 2022)

$Im(z)$ of $z = (5, 7) + (3, -2)$ is _____.

Select the correct option

◉	5 ✓
○	7
○	8
○	3

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:42:43 PM, 01 February 2022)

Sum of complex numbers (3 + 2i) and (1 + 7i) is _____.

Select the correct option

○	3+14i
◉	4 + 9i ✓
○	3+7i
○	3+9i

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Question # 3 of 5 (Start time: 10:42:59 PM, 01 February 2022) **Total Marks: 1**

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 then the $\lim_{z \to z_0} f(z)$ _____.

Select the correct option

Reload Math Equations

◉	does not exist
○	exists

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:43:26 PM, 01 February 2022)

If f(z)=z^2, then f(x+iy)= _____.

Select the correct option

◉	$x^2 - y^2 + 2xyi$ ✓
○	$x^2 - y^2 i - 2xyi$
○	$x^2 + y^2 - 2xyi$
○	$x^2 + y^2 i - 2xyi$

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Question # 2 of 5 (Start time: 11:14:26 PM, 01 February 2022)

Polar form of the complex number 2-2i is

Select the correct option

◉	$2\sqrt{2}cis\left(-\frac{\pi}{4}\right)$ ✓
○	none of these
○	$2cis\left(-\frac{\pi}{4}\right)$
○	$2cis\left(\frac{\pi}{4}\right)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 11:15:10 PM, 01 February 2022)

The associative law shows that _____.

Select the correct option

○	$z + w = z$
◉	$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ ✓
○	$z_1 (z_2 + z_3) = (z_1 z_2 + z_1 z_3)$
○	$z_1 + z_2 = z_2 + z_1$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) Quiz Start Time: 11:14 PM, 01 Februar

Question # 5 of 5 (Start time: 11:16:45 PM, 01 February 2022) Total M

A set S is said to be _____ set if for every pair of points z1, z2∈S there is a curve C joining them which is entirely contained in S.

Select the correct option

Reload Math Equat

- unbounded
- unconnected
- connected 
- bounded

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:15:04 PM, 01 February 2022)

A complex - valued function of the form $L(z) = Az + B$, *where* $A, B \in \mathbb{Z}$, represents a _____ transformation.

Select the correct option

- quadratic
- cubic
- none of the above
- linear ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 12:15:17 PM, 01 February 2022)

By definition, $\lim_{z \to z_0} f(z) = L$ means that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that _____.

Select the correct option

- $|f(z) - z_0| < \varepsilon$ whenever $0 < |z - L| < \delta$
- $|f(z) - z_0| < \delta$ whenever $0 < |z - L| < \varepsilon$
- $|f(z) - L| < \delta$ whenever $0 < |z - z_0| < \varepsilon$
- $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$ ✔

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 12:17:31 PM, 01 February 2022)

If two complex - valued functions f and g satisfy conditions $g(f(z)) = z$ and $f(g(w))$ then these functions are_____of each other.

Select the correct option

○	none of the above
○	conjugates
○	reciprocals
●	inverses

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 04:45:50 PM, 01 February 2022)

In an Argand diagram, the complex number z = 2i will be a point _____.

Select the correct option

○	2 units along the positive real axis
○	2 units along the positive imaginary axis ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 11:11:34 PM, 01 February 2022)

The principal argument of -5-5i is

Select the correct option

○	$\frac{-3\pi}{4}$
○ ✓	$\frac{5\pi}{4}$
○	$-\frac{5\pi}{4}$
○	$\frac{3\pi}{4}$

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ion # 4 of 5 (Start time: 10:18:01 PM, 01 February 2022)

Given a complex - valued function $s(t)$, a rotation by an angle $\frac{\pi}{3}$ is given by _____

the correct option

none of the above

$e^{\frac{2\pi}{T}}s(t)$

$e^{\frac{i\pi}{T}}s(t)$

$e^{\frac{3\pi}{T}}s(t)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:00:33 PM, 01 February 2022)

A complex-valued linear transformation is always _____ on the entire complex plane.

Select the correct option

- ○ none of the above
- ○ only onto
- ○ only one-to-one
- ○ one-to-one and onto ✓

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Question # 4 of 5 (Start time: 12:13:10 PM, 02 February 2022)

Which of the following does NOT describe the modulus of the complex number a + bi?

Select the correct option

◉	The difference between the numbers a and b. ✔
○	$\sqrt{a^2 + b^2}$
○	The magnitude of the vector representing the complex number a + bi.
○	The length of the directed line segment from the origin of the complex plane to the point (a, b).

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 07:59:20 PM, 01 February 2022)

According to _____, $(2 + i3) + (5 + i4) = (5 + i4) + (2 + i3)$

Select the correct option

○	Associative law
○	Commutative law ✓

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rH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

estion # 2 of 5 (Start time: 08:00:01 PM, 01 February 2022)

Which of the following statements is TRUE?

elect the correct option

○	The set Q of rational numbers is an extension of set Q' of irrational numbers.
○	The set R of real numbers is an extension of set C of complex numbers.
○	The set Q' of irrational numbers is an extension of set Z of integers.
○	The set C of complex numbers is an extension of set R of real numbers. ✓

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$$f(z) = \frac{1}{z + \bar{z}}$$

$z \neq \pm 2$

$z \neq \pm 1$

$z \neq \pm i$

$z \neq 0$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 12:12:35 PM, 02 February 2022)

$\sqrt{2}$ is a rational number.

Select the correct option

- (●) False
- ( ) True

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:11:12 PM, 02 February 2022)

Domain of the function

$$f(z) = \frac{1}{z^2 + 1}$$

is _____

Select the correct option

○	$z \neq \pm 2$
○	$z \neq \pm 1$
◉ ✓	$z \neq \pm i$
○	$z \neq 0$

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MTH632 QUIZ FILE

SOLVED BY SANAULLAH KHAN

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1) In general. 1^(1/n) is---------- $\exp\left(\frac{2k\pi}{n}i\right), k = 0,1,2,3.....,n-1$

2) the third root of 2i is ---------- $\sqrt[1]{2}\left[i\left(\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k \in z$

3) The distance b/t two complex numbers Z1 and Z2 is ---------- $|Z_1 - Z_2|$

4) The derivative of f at Z_0 i.e. $f' = \lim_{z\to z_0}\left(\frac{f(z)-f(z_0)}{z-z_0}\right)$ provided the limit exists. The function must be defined is a neighborhood of----------Z_0

5) A closed disc with center Z_0 and radius Ɛ is defined by---------- $|Z_1 - Z_0| \le \varepsilon$

6) Consider $\lim_{z\to z_0} g(z) = B.Then \lim_{z\to z_0}(f(z)g(z)) =$ ----------$A.B$

7) IF $f(z)=z^2$, $g(z)=1/z$, then g(f(3))=----------1/9

8)) The third cube roots of the number -8i are ---------- $2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k = 0,1,2$

9) If $f(z)=z^3$, $g(x)=z+2$, then g(f(5))=----------127

10) If $f(z)=z^3$, $g(x)=z+2$, then g(f(4))=----------66

11) If U and V are harmonic functions then f(Z)=U+iV is----------analytic func.

12) Harmonic conjugate of $u(x,y)=e^y consx$ is----------$e^y sinx+c$

13) A point at which a func. Ceases to be analytic is called a ---------point.----------
singular

14) A func. V is said to be harmonic if and only if----------
$V_{xx}+V_{yy}=0$

15) Consider $\lim_{z\to z_0} f(z) = A\, and \lim_{z\to z_0} g(z) = B.Then \lim_{z\to z_0}\frac{f(z)}{g(z)} =$ ----------$\frac{A}{B}, B \neq 0$

16) In the complex valued function Z=X-iY, the value of V_x=----------0

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17) If f(z)=X+aY+i(bX+cY) is analytic, then a,b,c equals to ----------c=1 and a=-b

18) The sequence $Z_n = -3 + i\frac{1}{n^3}(n = 1,2,3,.....)\ for\ n \to \infty$ **convergence to……-3**

19) The Cauchy-Riemann equation on a pair of real-valued functions of two real variables u(x,y) and v(x,y) are $U_x=V_y$ and----------
$U_y=-V_x$

20) If $e^{ax}\cos y$ **is harmonic then 'a' is----------.i**

21) If Cauchy-Riemann equations are not satisfied at a point Z_0 then func.Is not…….at Z_0----------
differentiable

22) In the complex valued func. Z=x-iy, the value of U_x=----------1

23) In the complex valued func. Z=x-iy, the value of V_x=----------0

24)Consider $\lim_{z\to z_0} f(z) = A\ and\ \lim_{z\to z_0} g(z) = B.Then\ \lim_{z\to z_0}(f(z) \pm g(z)) = \text{---------}\ A \pm B$

25) A func. F(z) is analytic func. If ……part f(z) is analytic----------both (a) and (b)

26) Any polynomial $p(z) = a_0 + a_1 z + a_2 z^2 + ... + a_n z^n (a_n \neq 0)$ **of degree** $n(n \geq 1)$ **has least one zero, i.e. there exists at least……Z_0 such that P(Z_0)=0----------one pint**

27) Each simple closed curve C divides the plane into two domains(Connected Open Sets). One of the which is unbounded and is called……of C----------Exterior

28) A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end----------
True

29) A domains(Connected Open Sets) is called……connected if interior of every simple closed contour C is contained in D.----------simply

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30) if we write $\Delta_z = z - z_0$ **then the func.** $f'(z_0) = \lim_{z \to z_0}\left(\frac{f(z)-f(z_0)}{z-z_0}\right)$ **can be expressed as** $f'(z_0) =$ ---------- $\lim_{\Delta z \to 0}\left(\frac{f(z_0-\Delta_z)-f(z_0)}{\Delta_z}\right)$.

31) If f(z)=u+iv in polar form is analytic then $\frac{\partial u}{\partial r}$ **is----------** $\frac{1}{r}\frac{\partial v}{\partial \theta}$.

32) If f(z)=u+iv in polar form is analytic then $\frac{\partial u}{\partial \theta}$ **is----------** $-r\frac{\partial v}{\partial r}$.

33) A func. v is called a conjugate harmonic func. For a harmonic func. u in Ω **ehenever----------f=u+iv is analytic.**

34) The func. $f(x+iy) = x^3 + ax^2y + bxy^2 + cy^3$ **is analytic if----------a=3i,b=-3 and c=i.**

35) There exist no analytic func. f such that----------Re f(z)=y^2-2x

36) The harmonic conjugate of u(x,y)=2x(1-y) is---------- $x^2 - y^2 + 2y + c$

37) The harmonic conjugate of $2x - x^3 + 3xy^2$ **is----------** $2y - 3x^2y + y^3$

38) If the real part of an analytic func. f(z) is x^2 -y^2-y,then imaginary part is---------- 2xy+x.

39)) If the imaginary part of an analytic func. f(z) is 2xy+y,then real part is---------- x^2 **- y^2+x**

40) $f(z) = \overline{z}$ **is differentiable----------nowhere.**

41) $f(z) = |\overline{z}|^2$ **is differentiable----------only at z=0 (but not analytic at z=0).**

42) If
$$f(z) = \begin{cases} \frac{xy}{x^2+y^2}, & if \ z \neq 0 \\ 0 & if \ z = 0 \end{cases}$$
then f(z) is----------not diff. at z=0

43) $f(z) = e^z$ **is analytic----------everywhere.**

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44) $e^x(\cos y - i\sin y)$ is----------not analytic.

45) If f(z) is analytic then $\overline{f(\bar{z})}$ is----------analytic.

46) The point at which $f(z)=\dfrac{z^2-z}{z^2-3z+1}$ is not analytic are---------- i and $-i$

47) The harmonic conjugate of $u=\log\sqrt{x^2+y^2}$ is---------- $\tan^{-1}\left(\dfrac{y}{x}\right)$.

48) If f(z)=z(2-z) then f(1+i)=----------i.

49) If $f(z)=|\bar{z}|$ then f(3-4i)=----------5

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